

SOLIDWORKS API (.NET VBA C#) — Reference

(127/337): IFreePointCurveFeatureData- IGroundPlaneFeatureData

SOLIDWORKS API Documentation (converted from CHM)

IFreePointCurveFeatureData (Type)

IFreePointCurveFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFreePointCurveFeatureData Interface

Allows access to a curve created using X, Y, Z coordinates for the points.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IFreePointCurveFeatureData
```

Visual Basic (Usage)



```
Dim instance As IFreePointCurveFeatureData
```

C#

```
public interface IFreePointCurveFeatureData
```

C++/CLI

```
public interface class IFreePointCurveFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [FreePointCurveFeatureData](#).

Example

[Insert Free Point Curve Feature \(C#\)](#)
[Insert Free Point Curve Feature \(VB.NET\)](#)
[Insert Free Point Curve Feature \(VBA\)](#)

Remarks

See SOLIDWORKS Help for additional information about curves through free points.

Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)

Access Diagram

[FreePointCurveFeatureData](#)

See Also

[IFreePointCurveFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IFreePointCurveFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFreePointCurveFeatureData Interface

The following tables list the members exposed by [IFreePointCurveFeatureData](#).

Public Properties

Name	Description
 PointArray	Gets or sets the list of X, Y, Z coordinates for the points for this free-point curve.

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Public Methods

Name	Description
 GetPointCount	Gets the number of doubles (3 * number of points) of this free-point curve.
 IGetPointArray	Gets the list of X, Y, Z coordinates for the points for this free-point curve.
 ISetPointArray	Sets the list of X, Y, Z coordinates for the points for this free-point curve.
 LoadPointsFromFile	Loads the points for this free-point curve from a file.
 SavePointsToFile	Saves the points for this free-point curve to a file.

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See Also

[IFreePointCurveFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IFreePointCurveFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFreePointCurveFeatureData Interface

For a list of all members of this type, see [IFreePointCurveFeatureData members](#).

Public Methods

Name	Description
 GetPointCount	Gets the number of doubles (3 * number of points) of this free-point curve.
 IGetPointArray	Gets the list of X, Y, Z coordinates for the points for this free-point curve.
 ISetPointArray	Sets the list of X, Y, Z coordinates for the points for this free-point curve.
 LoadPointsFromFile	Loads the points for this free-point curve from a file.
 SavePointsToFile	Saves the points for this free-point curve to a file.

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See Also

[IFreePointCurveFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IFreePointCurveFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFreePointCurveFeatureData Interface

For a list of all members of this type, see [IFreePointCurveFeatureData members](#).

Public Properties

Name	Description
 PointArray	Gets or sets the list of X, Y, Z coordinates for the points for this free-point curve.

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See Also

[IFreePointCurveFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

GetPointCount Method (IFreePointCurveFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFreePointCurveFeatureData Interface](#) : GetPointCount Method (IFreePointCurveFeatureData)

Gets the number of doubles (3 * number of points) of this free-point curve.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPointCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFreePointCurveFeatureData  
Dim value As System.Integer
```

```
value = instance.GetPointCount()
```

C#

```
System.int GetPointCount()
```

C++/CLI

```
System.int GetPointCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of doubles (3 * number of points) of this free-point curve

Visual Basic for Applications (VBA) Syntax

See [FreePointCurveFeatureData::GetPointCount](#).

Example

[Insert Free Point Curve Feature \(C#\)](#)
[Insert Free Point Curve Feature \(VB.NET\)](#)
[Insert Free Point Curve Feature \(VBA\)](#)

See Also

[IFreePointCurveFeatureData Interface](#)
[IFreePointCurveFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

IGetPointArray Method (IFreePointCurveFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFreePointCurveFeatureData Interface](#) : IGetPointArray Method (IFreePointCurveFeatureData)

Gets the list of X, Y, Z coordinates for the points for this free-point curve.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPointArray() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFreePointCurveFeatureData
Dim value As System.Double

value = instance.IGetPointArray()
```

C#

```
System.double IGetPointArray()
```

C++/CLI

```
System.double IGetPointArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles containing 3 * points (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

The ArrayOut argument is an array of doubles containing 3 * points:

[x1, y1, z1, x2, y2, z2, ...]

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[IFreePointCurveFeatureData Interface](#)
[IFreePointCurveFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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ISetPointArray Method (IFreePointCurveFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFreePointCurveFeatureData Interface](#) : ISetPointArray Method (IFreePointCurveFeatureData)

Sets the list of X, Y, Z coordinates for the points for this free-point curve.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPointArray( _
    ByVal FeatCount As System.Integer, _
    ByRef ArrayDataIn As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As IFreePointCurveFeatureData
Dim FeatCount As System.Integer
Dim ArrayDataIn As System.Double
```

```
instance.ISetPointArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetPointArray(
    System.int FeatCount,
    ref System.double ArrayDataIn
)
```

C++/CLI

```
void ISetPointArray(
    & System.int FeatCount,
    & System.double% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of features

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of doubles containing 3 * points (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [in-process Methods](#) for details about this type of method.

Remarks

The *ArrayDataIn* argument is an array of doubles containing 3 * points:

[x1, y1, z1, x2, y2, z2, ...]

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[IFreePointCurveFeatureData Interface](#)
[IFreePointCurveFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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LoadPointsFromFile Method (IFreePointCurveFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFreePointCurveFeatureData Interface](#) : LoadPointsFromFile Method (IFreePointCurveFeatureData)

Loads the points for this free-point curve from a file.

.NET Syntax

Visual Basic (Declaration)

```
Function LoadPointsFromFile( _
    ByVal FileName As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFreePointCurveFeatureData
Dim FileName As System.String
Dim value As System.Boolean

value = instance.LoadPointsFromFile(FileName)
```

C#

```
System.bool LoadPointsFromFile(  
    System.string FileName  
)
```

C++/CLI

```
System.bool LoadPointsFromFile(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path and file name of the free-point curve file that contains the points to load; you can specify **.sldcrv** files or **.txt** files that use the same format as **.sldcrv** files

Return Value

True if the points loaded successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [FreePointCurveFeatureData::LoadPointsFromFile](#).

Remarks

See SOLIDWORKS Help for details about free-point curve files.

See Also

[IFreePointCurveFeatureData Interface](#)
[IFreePointCurveFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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PointArray Property (IFreePointCurveFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFreePointCurveFeatureData Interface](#) : PointArray Property (IFreePointCurveFeatureData)

Gets or sets the list of X, Y, Z coordinates for the points for this free-point curve.

.NET Syntax

Visual Basic (Declaration)

```
Property PointArray As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFreePointCurveFeatureData  
Dim value As System.Object
```

```
instance.PointArray = value
```

```
value = instance.PointArray
```

C#

```
System.object PointArray {get; set;}
```

C++/CLI

```
property System.Object^ PointArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles containing 3 * points (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [FreePointCurveFeatureData::PointArray](#).

Example

[Insert Free Point Curve Feature \(C#\)](#)
[Insert Free Point Curve Feature \(VB.NET\)](#)
[Insert Free Point Curve Feature \(VBA\)](#)

Remarks

The ArrayDataIn argument is an array of doubles containing 3 * points:

[x1, y1, z1, x2, y2, z2, ...]

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[IFreePointCurveFeatureData Interface](#)
[IFreePointCurveFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SavePointsToFile Method (IFreePointCurveFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFreePointCurveFeatureData Interface](#) : SavePointsToFile Method (IFreePointCurveFeatureData)

Saves the points for this free-point curve to a file.

.NET Syntax

Visual Basic (Declaration)

```
Function SavePointsToFile( _
    ByVal FileName As System.String _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IFreePointCurveFeatureData
Dim FileName As System.String
Dim value As System.Boolean

value = instance.SavePointsToFile(FileName)
```

C#

```
System.bool SavePointsToFile(
    System.string FileName
)
```

C++/CLI



```
System.bool SavePointsToFile(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path and file name of the file to which to save the points; you can specify **.sldcrv** files or **.txt** as the filename extension

Return Value

True if the points are saved to the file, false if not

Visual Basic for Applications (VBA) Syntax

See [FreePointCurveFeatureData::SavePointsToFile](#).

Example

[Insert Free Point Curve Feature \(C#\)](#)
[Insert Free Point Curve Feature \(VB.NET\)](#)
[Insert Free Point Curve Feature \(VBA\)](#)

Remarks

Exported units, e.g., meters, of the document that owns the feature, and not necessarily those of the active document, are used when data is saved to a file.

See SOLIDWORKS Help for details about free-point curve files.

See Also

[IFreePointCurveFeatureData Interface](#)
[IFreePointCurveFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

IGearMateFeatureData (Type)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGearMateFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGearMateFeatureData Interface

Allows access to gear mate features.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface IGearMateFeatureData

Visual Basic (Usage)



Dim instance As [IGearMateFeatureData](#)

C#

public interface IGearMateFeatureData

C++/CLI

public interface class IGearMateFeatureData

Visual Basic for Applications (VBA) Syntax

See [GearMateFeatureData](#).

Example

'VBA

' *****

' 1. Open *public_documents\samples\tutorial\api\MechanicalMates\spurgear.sldasm*.

' 2. Delete **GearMate1** from the Mates folder of the FeatureManager design tree.

' 3. Run the macro.

' 4. Inspect the Mates folder in the FeatureManager design tree.

' *****

Dim swApp As SldWorks.SldWorks

Dim Part As SldWorks.ModelDoc2

Dim boolstatus As Boolean

Option Explicit

Sub main()

Set swApp = Application.SldWorks

Set Part = swApp.ActiveDoc

boolstatus = Part.Extension.SelectByRay(-5.60801689982782E-03, -3.24062886681986E-03, -2.52602902980925E-03,
-0.577381545199981, -0.577287712085548, -0.577381545199979, 5.93757005519753E-04, 2, True, 1, 0)

boolstatus = Part.Extension.SelectByRay(3.67362652137331E-02, -6.55599730123413E-04, -2.90761848191323E-03,
-0.577381545199981, -0.577287712085548, -0.577381545199979, 5.93757005519753E-04, 2, True, 1, 0)

' Create GearMateFeatureData

Dim MateData As SldWorks.GearMateFeatureData

Set MateData = Part.**CreateMateData**(10)

' Set the entities To mate

Dim EntitiesToMate(1) As Object

Set EntitiesToMate(0) = Part.SelectionManager.GetSelectedObject6(1, -1)

Set EntitiesToMate(1) = Part.SelectionManager.GetSelectedObject6(2, -1)

Dim EntitiesToMateVar As Variant

EntitiesToMateVar = EntitiesToMate

MateData.**EntitiesToMate** = (EntitiesToMateVar)

' Set the gear ratio numerator

MateData.**GearRatioNumerator** = 0.012954

```
' Set the gear ratio denominator
MateData.GearRatioDenominator = 0.012954

' Set the mate orientation direction
MateData.Reverse = False

' Create the mate
Dim MateFeature As SldWorks.Feature
Set MateFeature = Part.CreateMate(MateData)
Part.ClearSelection2 True
Part.EditRebuild3

End Sub
```

 Example

[Create Gear Mate \(C#\)](#)

 Remarks

A gear mate forces two components to rotate relative to one another about selected axes.

[IMateFeatureData](#) is the parent of this mate interface.

To create a gear mate:

1. Follow general instructions in the [IAssemblyDoc::CreateMate](#) Remarks.
2. Specify [IGearMateFeatureData.EntitiesToMate](#) or use [IModelDocExtension::SelectByID2](#) to pre-select the entities using Mark = 1.
3. Specify other properties of the GearMateFeatureData object.

To edit a gear mate:

1. Access its feature and call [IFeature::GetDefinition](#) to get the MateFeatureData object.
2. Cast the MateFeatureData object to a GearMateFeatureData object.
3. Modify the GearMateFeatureData object.
4. Call [IFeature::ModifyDefinition](#).

To delete a gear mate, use [IModelDocExtension::DeleteSelection2](#).

 Access Diagram

[GearMateFeatureData](#)

 See Also

[IGearMateFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IGearMateFeatureData Interface Members

[See Also](#) [Properties](#)



| [SolidWorks.Interop.sldworks Namespace](#) : IGearMateFeatureData Interface |

The following tables list the members exposed by [IGearMateFeatureData](#).

 Public Properties

Name	Description
 EntitiesToMate	Gets or sets the entities to mate in this gear mate.
 GearRatioDenominator	Gets or sets the denominator of the gear ratio of this gear mate.
 GearRatioNumerator	Gets or sets the numerator of the gear ratio of this gear mate.
 Reverse	Gets or sets whether to change the direction of rotation of the gears relative to one another in this gear mate.

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 See Also

[IGearMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

IGearMateFeatureData Interface Properties

[See Also](#)



| [SolidWorks.Interop.sldworks Namespace](#) : IGearMateFeatureData Interface |

For a list of all members of this type, see [IGearMateFeatureData members](#).

Public Properties

Name	Description
 EntitiesToMate	Gets or sets the entities to mate in this gear mate.
 GearRatioDenominator	Gets or sets the denominator of the gear ratio of this gear mate.
 GearRatioNumerator	Gets or sets the numerator of the gear ratio of this gear mate.
 Reverse	Gets or sets whether to change the direction of rotation of the gears relative to one another in this gear mate.

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See Also

[IGearMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

[Send comments on this topic.](#)

SOLIDWORKS API Help

EntitiesToMate Property (IGearMateFeatureData)

[See Also](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IGearMateFeatureData Interface](#) : EntitiesToMate Property (IGearMateFeatureData)

Gets or sets the entities to mate in this gear mate.

.NET Syntax

Visual Basic (Declaration)

Property EntitiesToMate As System.Object

Visual Basic (Usage)

 Copy Code

```
Dim instance As IGearMateFeatureData
Dim value As System.Object

instance.EntitiesToMate = value

value = instance.EntitiesToMate
```

C#

```
System.object EntitiesToMate {get; set;}
```

C++/CLI

```
property System.Object^ EntitiesToMate {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of two entities to mate (cylindrical or conical [faces](#), [axes](#), or linear [edges](#))

Visual Basic for Applications (VBA) Syntax

See [GearMateFeatureData::EntitiesToMate](#).

Example

See the [IGearMateFeatureData](#) example.

Remarks

Instead of specifying this property, you can use [IModelDocExtension::SelectByID2](#) to pre-select the entities to mate using Mark = 1. You can pre select mate entities during mate creation, but not during mate editing.

See Also

[IGearMateFeatureData Interface](#)
[IGearMateFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GearRatioDenominator Property (IGearMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGearMateFeatureData Interface](#) : GearRatioDenominator Property (IGearMateFeatureData)

Gets or sets the denominator of the gear ratio of this gear mate.

.NET Syntax

Visual Basic (Declaration)

```
Property GearRatioDenominator As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGearMateFeatureData  
Dim value As System.Double
```

```
instance.GearRatioDenominator = value
```

```
value = instance.GearRatioDenominator
```

C#

```
System.double GearRatioDenominator {get; set;}
```

C++/CLI

```
property System.double GearRatioDenominator {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Gear ratio denominator

Visual Basic for Applications (VBA) Syntax

See [GearMateFeatureData::GearRatioDenominator](#).

Example

See the [IGearMateFeatureData](#) example.

See Also

[IGearMateFeatureData Interface](#)
[IGearMateFeatureData Members](#)
[IGearMateFeatureData::GearRatioNumerator Property \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

GearRatioNumerator Property (IGearMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGearMateFeatureData Interface](#) : GearRatioNumerator Property (IGearMateFeatureData)

Gets or sets the numerator of the gear ratio of this gear mate.

.NET Syntax

Visual Basic (Declaration)

Property GearRatioNumerator As System.Double

Visual Basic (Usage)



```
Dim instance As IGearMateFeatureData  
Dim value As System.Double
```

```
instance.GearRatioNumerator = value
```

```
value = instance.GearRatioNumerator
```

C#

```
System.double GearRatioNumerator {get; set;}
```

C++/CLI

```
property System.double GearRatioNumerator {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Gear ratio numerator

Visual Basic for Applications (VBA) Syntax

See [GearMateFeatureData::GearRatioNumerator](#).

Example

See the [IGearMateFeatureData](#) example.

See Also

[IGearMateFeatureData Interface](#)
[IGearMateFeatureData Members](#)
[IGearMateFeatureData::GearRatioDenominator Property \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

Reverse Property (IGearMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGearMateFeatureData Interface](#) : Reverse Property (IGearMateFeatureData)

Gets or sets whether to change the direction of rotation of the gears relative to one another in this gear mate.

.NET Syntax

Visual Basic (Declaration)

Property Reverse As System.Boolean

Visual Basic (Usage)



```
Dim instance As IGearMateFeatureData  
Dim value As System.Boolean
```

```
instance.Reverse = value
```

```
value = instance.Reverse
```

C#

```
System.bool Reverse {get; set;}
```

C++/CLI

```
property System.bool Reverse {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse the direction of rotation, false to not

Visual Basic for Applications (VBA) Syntax

See [GearMateFeatureData::Reverse](#).

Example

See the [IGearMateFeatureData](#) example.

See Also

[IGearMateFeatureData Interface](#)
[IGearMateFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

IGeneralTableAnnotation (Type)

IGeneralTableAnnotation Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralTableAnnotation Interface

Allows access to the general table annotation.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IGeneralTableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IGeneralTableAnnotation
```

C#

```
public interface IGeneralTableAnnotation
```

C++/CLI

```
public interface class IGeneralTableAnnotation
```

Visual Basic for Applications (VBA) Syntax

See [GeneralTableAnnotation](#).

Example

[Sort Table \(C#\)](#)
[Sort Table \(VB.NET\)](#)
[Sort Table \(VBA\)](#)

Accessors

[IGeneralTableFeature::GetTableAnnotations](#) and [IGeneralTableFeature::IGetTableAnnotations](#)

Access Diagram

[GeneralTableAnnotation](#)

See Also

[IGeneralTableAnnotation Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IGeneralTableAnnotation Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralTableAnnotation Interface

For a list of all members of this type, see [IGeneralTableAnnotation members](#).

Public Methods

Name	Description
Sort	Sorts this table by the specified column in the specified sorting direction.

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See Also

[IGeneralTableAnnotation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGeneralTableAnnotation Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralTableAnnotation Interface

The following tables list the members exposed by [IGeneralTableAnnotation](#).

Public Properties

Name	Description
 GeneralTable	Gets the general table feature for this annotation.

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Public Methods

Name	Description
 Sort	Sorts this table by the specified column in the specified sorting direction.

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See Also

[IGeneralTableAnnotation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGeneralTableAnnotation Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralTableAnnotation Interface

For a list of all members of this type, see [IGeneralTableAnnotation members](#).

Public Properties

Name	Description
 GeneralTable	Gets the general table feature for this annotation.

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See Also

[IGeneralTableAnnotation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GeneralTable Property (IGeneralTableAnnotation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralTableAnnotation Interface](#) : GeneralTable Property (IGeneralTableAnnotation)

Gets the general table feature for this annotation.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property GeneralTable As GeneralTableFeature

Visual Basic (Usage)



```
Dim instance As IGeneralTableAnnotation  
Dim value As GeneralTableFeature
```

```
value = instance.GeneralTable
```

C#

```
GeneralTableFeature GeneralTable {get;}
```

C++/CLI

```
property GeneralTableFeature^ GeneralTable {  
    GeneralTableFeature^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[IGeneralTableFeature](#)

Visual Basic for Applications (VBA) Syntax

See [GeneralTableAnnotation::GeneralTable](#).

See Also

[IGeneralTableAnnotation Interface](#)
[IGeneralTableAnnotation Members](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Sort Method (IGeneralTableAnnotation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralTableAnnotation Interface](#) : Sort Method (IGeneralTableAnnotation)

Sorts this table by the specified column in the specified sorting direction.

.NET Syntax

Visual Basic (Declaration)

```
Function Sort(  
    ByVal ColumnIndex As System.Integer, _  
    ByVal SortAscending As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGeneralTableAnnotation  
Dim ColumnIndex As System.Integer  
Dim SortAscending As System.Boolean  
Dim value As System.Boolean  
  
value = instance.Sort(ColumnIndex, SortAscending)
```

C#

```
System.bool Sort(  
    System.int ColumnIndex,  
    System.bool SortAscending  
)
```

C++/CLI

```
System.bool Sort(  
& System.int ColumnIndex,
```

```
& System.bool SortAscending  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ColumnIndex

0-based index of column to sort by

SortAscending

True to sort ascending, false to sort descending

Return Value

True if sorted successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [GeneralTableAnnotation::Sort](#).

Example

[Sort Table \(VBA\)](#)

[Sort Table \(VB.NET\)](#)

[Sort Table \(C#\)](#)

Remarks

See [Sorting Tables](#) for more information.

See Also

[IGeneralTableAnnotation Interface](#)

[IGeneralTableAnnotation Members](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

IGeneralTableFeature (Type)

IGeneralTableFeature Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralTableFeature Interface

Allows access to the general table feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface IGeneralTableFeature

Visual Basic (Usage) [Copy Code](#)

Dim instance As [IGeneralTableFeature](#)

C#

public interface IGeneralTableFeature

C++/CLI

public interface class IGeneralTableFeature

Visual Basic for Applications (VBA) Syntax

See [GeneralTableFeature](#).

Example

[Hide and Show Row \(VBA\)](#)
[Get General Table Feature \(C#\)](#)
[Get General Table Feature \(VB.NET\)](#)
[Get General Table Feature \(VBA\)](#)

Remarks

You can get this object using:

- [ISelectionMgr::GetSelectedObject6](#) when a general table feature is selected in the FeatureManager design tree of the drawing.
- [ITableAnnotation::GeneralTableFeature](#).
- [IGeneralTableAnnotation::GeneralTable](#).

The filename extension for a general table is **.sldtbt**. Table templates are not supplied for general tables.

Accessors

See **Remarks**.

Access Diagram

[GeneralTableFeature](#)

See Also

[IGeneralTableFeature Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGeneralToleranceTableFeature Interface](#)

IGeneralTableFeature Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralTableFeature Interface

For a list of all members of this type, see [IGeneralTableFeature members](#).

Public Methods

Name	Description
GetFeature	Gets the general table feature.
GetTableAnnotationCount	Gets the number of table annotations for this general table feature.
GetTableAnnotations	Gets the table annotations for this general table feature.
IGetTableAnnotations	Gets the table annotations for this general table feature.

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See Also

[IGeneralTableFeature Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGeneralToleranceTableFeature Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGeneralTableFeature Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralTableFeature Interface

The following tables list the members exposed by [IGeneralTableFeature](#).

Public Methods

Name	Description
GetFeature	Gets the general table feature.
GetTableAnnotationCount	Gets the number of table annotations for this general table feature.
GetTableAnnotations	Gets the table annotations for this general table feature.
IGetTableAnnotations	Gets the table annotations for this general table feature.

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See Also

[IGeneralTableFeature Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGeneralToleranceTableFeature Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeature Method (IGeneralTableFeature)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralTableFeature Interface](#) : GetFeature Method (IGeneralTableFeature)

Gets the general table feature.

.NET Syntax

Visual Basic (Declaration)

Function GetFeature() As Feature

Visual Basic (Usage)



```
Dim instance As IGeneralTableFeature  
Dim value As Feature  
  
value = instance.GetFeature()
```

C#

```
Feature GetFeature()
```

C++/CLI

```
Feature^ GetFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [GeneralTableFeature::GetFeature](#).

Example

[Get General Table Feature \(C#\)](#)
[Get General Table Feature \(VB.NET\)](#)
[Get General Table Feature \(VBA\)](#)

See Also

[IGeneralTableFeature Interface](#)
[IGeneralTableFeature Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTableAnnotationCount Method (IGeneralTableFeature)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralTableFeature Interface](#) : GetTableAnnotationCount Method (IGeneralTableFeature)

Gets the number of table annotations for this general table feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTableAnnotationCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGeneralTableFeature  
Dim value As System.Integer  
  
value = instance.GetTableAnnotationCount()
```

C#

```
System.int GetTableAnnotationCount()
```

C++/CLI

```
System.int GetTableAnnotationCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of table annotations

Visual Basic for Applications (VBA) Syntax

See [GeneralTableFeature::GetTableAnnotationCount](#).

Example

[Hide and Show Row \(VBA\)](#)
[Get General Table Feature \(C#\)](#)
[Get General Table Feature \(VB.NET\)](#)
[Get General Table Feature \(VBA\)](#)

Remarks

Call this method before calling [IGeneralTableFeature::IGetTableAnnotations](#) to get the size of the array.

See Also

[IGeneralTableFeature Interface](#)
[IGeneralTableFeature Members](#)
[IGeneralTableFeature::GetTableAnnotations Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTableAnnotations Method (IGeneralTableFeature)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralTableFeature Interface](#) : GetTableAnnotations Method (IGeneralTableFeature)

Gets the table annotations for this general table feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTableAnnotations() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGeneralTableFeature  
Dim value As System.Object  
  
value = instance.GetTableAnnotations()
```

C#

```
System.object GetTableAnnotations()
```

C++/CLI

```
System.Object^ GetTableAnnotations();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [general table annotations](#)

Visual Basic for Applications (VBA) Syntax

See [GeneralTableFeature::GetTableAnnotations](#).

Example

[Hide and Show Row \(VBA\)](#)
[Get General Table Feature \(C#\)](#)
[Get General Table Feature \(VB.NET\)](#)
[Get General Table Feature \(VBA\)](#)

See Also

[IGeneralTableFeature Interface](#)
[IGeneralTableFeature Members](#)
[IGeneralTableFeature::IGetTableAnnotations Method](#)
[IGeneralTableFeature::GetTableAnnotationCount Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTableAnnotations Method (IGeneralTableFeature)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralTableFeature Interface](#) : IGetTableAnnotations Method (IGeneralTableFeature)

Gets the table annotations for this general table feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTableAnnotations( _  
    ByVal Count As System.Integer _  
) As TableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IGeneralTableFeature  
Dim Count As System.Integer  
Dim value As TableAnnotation  
  
value = instance.IGetTableAnnotations(Count)
```

C#

```
TableAnnotation IGetTableAnnotations(  
    System.int Count  
)
```

C++/CLI

```
TableAnnotation^ IGetTableAnnotations(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of table annotations

Return Value

- in-process, unmanaged C++: Pointer to an array of [IGeneralTableAnnotation](#) objects
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Call [IGeneralTableFeature::GetTableAnnotationCount](#) before calling this method to get the size of the array.

See Also

[IGeneralTableFeature Interface](#)
[IGeneralTableFeature Members](#)
[IGeneralTableFeature::GetTableAnnotations Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

IGeneralToleranceTableAnnotation (Type)

IGeneralToleranceTableAnnotation Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralToleranceTableAnnotation Interface

Allows access to the general tolerance table annotation.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IGeneralToleranceTableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IGeneralToleranceTableAnnotation
```

C#

```
public interface IGeneralToleranceTableAnnotation
```

C++/CLI

```
public interface class IGeneralToleranceTableAnnotation
```

Visual Basic for Applications (VBA) Syntax

See [GeneralToleranceTableAnnotation](#).

Example

[Insert General Tolerance Table \(VBA\)](#)
[Insert General Tolerance Table \(VB.NET\)](#)
[Insert General Tolerance Table \(C#\)](#)

Accessors

[IModelDocExtension::InsertGeneralToleranceTableAnnotation](#)
[IGeneralToleranceTableFeature::GetTableAnnotations](#)

Access Diagram

[GeneralToleranceTableAnnotation](#)

See Also

[IGeneralToleranceTableAnnotation Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IGeneralToleranceTableAnnotation Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralToleranceTableAnnotation Interface

The following tables list the members exposed by [IGeneralToleranceTableAnnotation](#).

Public Properties

Name	Description
 GeneralToleranceTableFeature	Gets the general tolerance table feature for this annotation.

[Top](#)

See Also

[IGeneralToleranceTableAnnotation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGeneralToleranceTableAnnotation Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IGeneralToleranceTableAnnotation Interface

For a list of all members of this type, see [IGeneralToleranceTableAnnotation members](#).

Public Properties

Name	Description
 GeneralToleranceTableFeature	Gets the general tolerance table feature for this annotation.

[Top](#)

See Also

[IGeneralToleranceTableAnnotation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GeneralToleranceTableFeature Property (IGeneralToleranceTableAnnotation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralToleranceTableAnnotation Interface](#) : GeneralToleranceTableFeature Property (IGeneralToleranceTableAnnotation)

Gets the general tolerance table feature for this annotation.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property GeneralToleranceTableFeature As GeneralToleranceTableFeature
```

Visual Basic (Usage)



```
Dim instance As IGeneralToleranceTableAnnotation  
Dim value As GeneralToleranceTableFeature  
  
value = instance.GeneralToleranceTableFeature
```

C#

```
GeneralToleranceTableFeature GeneralToleranceTableFeature {get;}
```

C++/CLI

```
property GeneralToleranceTableFeature^ GeneralToleranceTableFeature {  
    GeneralToleranceTableFeature^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[IGeneralToleranceTableFeature](#)

Visual Basic for Applications (VBA) Syntax

See [GeneralToleranceTableAnnotation::GeneralToleranceTableFeature](#).

Example

See the [IGeneralToleranceTableAnnotation](#) examples.

See Also

[IGeneralToleranceTableAnnotation Interface](#)
[IGeneralToleranceTableAnnotation Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

IGeneralToleranceTableFeature (Type)

IGeneralToleranceTableFeature Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralToleranceTableFeature Interface

Allows access to the general tolerance table feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IGeneralToleranceTableFeature
```

Visual Basic (Usage)

```
Dim instance As IGeneralToleranceTableFeature
```

C#

```
public interface IGeneralToleranceTableFeature
```

C++/CLI

```
public interface class IGeneralToleranceTableFeature
```

Visual Basic for Applications (VBA) Syntax

See [GeneralToleranceTableFeature](#).

Example

[Insert General Tolerance Table \(VBA\)](#)
[Insert General Tolerance Table \(VB.NET\)](#)
[Insert General Tolerance Table \(C#\)](#)

Remarks

You can access this object by calling:

- [ISelectionMgr::GetSelectedObject6](#) after selecting a general tolerance table feature in the Tables folder of the FeatureManager design tree
- or -
- [IGeneralToleranceTableAnnotation::GeneralToleranceTableFeature](#)

Accessors

See [Remarks](#).

Access Diagram

[GeneralToleranceTableFeature](#)

See Also

[IGeneralToleranceTableFeature Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGeneralTableFeature Interface](#)
[IModelDocExtension::InsertGeneralToleranceTableAnnotation Method \(\)](#)

IGeneralToleranceTableFeature Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralToleranceTableFeature Interface

For a list of all members of this type, see [IGeneralToleranceTableFeature members](#).

Public Methods

Name	Description
 GetFeature	Gets the general tolerance table feature.
 GetTableAnnotations	Gets the table annotations for this general tolerance table feature.

[Top](#)

See Also

[IGeneralToleranceTableFeature Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGeneralTableFeature Interface](#)
[IModelDocExtension::InsertGeneralToleranceTableAnnotation Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGeneralToleranceTableFeature Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGeneralToleranceTableFeature Interface

The following tables list the members exposed by [IGeneralToleranceTableFeature](#).

Public Methods

Name	Description
 GetFeature	Gets the general tolerance table feature.
 GetTableAnnotations	Gets the table annotations for this general tolerance table feature.

[Top](#)

See Also

[IGeneralToleranceTableFeature Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGeneralTableFeature Interface](#)
[IModelDocExtension::InsertGeneralToleranceTableAnnotation Method \(\)](#)

SOLIDWORKS API Help

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GetFeature Method (IGeneralToleranceTableFeature)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralToleranceTableFeature Interface](#) : GetFeature Method (IGeneralToleranceTableFeature)

Gets the general tolerance table feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As IGeneralToleranceTableFeature  
Dim value As Feature  
  
value = instance.GetFeature()
```

C#

```
Feature GetFeature()
```

C++/CLI

```
Feature^ GetFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [GeneralToleranceTableFeature::GetFeature](#).

Example

See the [IGeneralToleranceTableFeature](#) examples.

See Also

[IGeneralToleranceTableFeature Interface](#)
[IGeneralToleranceTableFeature Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTableAnnotations Method (IGeneralToleranceTableFeature)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGeneralToleranceTableFeature Interface](#) : GetTableAnnotations Method (IGeneralToleranceTableFeature)

Gets the table annotations for this general tolerance table feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTableAnnotations() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGeneralToleranceTableFeature  
Dim value As System.Object  
  
value = instance.GetTableAnnotations()
```

C#

```
System.object GetTableAnnotations()
```

C++/CLI

```
System.Object^ GetTableAnnotations();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IGeneralToleranceTableAnnotations](#)

Visual Basic for Applications (VBA) Syntax

See [GeneralToleranceTableFeature::GetTableAnnotations](#).

See Also

[IGeneralToleranceTableFeature Interface](#)
[IGeneralToleranceTableFeature Members](#)

Availability

IGraphicsBody (Type)

IGraphicsBody Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGraphicsBody Interface

Allows access to a graphics body.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IGraphicsBody
```

Visual Basic (Usage)



```
Dim instance As IGraphicsBody
```

C#

```
public interface IGraphicsBody
```

C++/CLI

```
public interface class IGraphicsBody
```

Visual Basic for Applications (VBA) Syntax

See [GraphicsBody](#).

Example

See the [IFacet](#) examples.

Remarks

For more information about graphics bodies, read the **Graphics Mesh and Mesh BREP Bodies** topic in **SOLIDWORKS Design user-interface help > Parts and Features**.

Accessors

[IBody2::GetGraphicsBody](#)

Access Diagram

[GraphicsBody](#)

See Also

[IGraphicsBody Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMeshBody Interface](#)

IGraphicsBody Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGraphicsBody Interface

The following tables list the members exposed by [IGraphicsBody](#).

Public Methods

Name	Description
------	-------------

 GetBody	Gets the body associated with this graphics body.
 GetFacetCount	Gets the number of facets on this graphics body.
 GetFacets	Gets the facets on this graphics body.
 SurfaceFromMesh	Creates a surface from the selected facets in this graphics body.

[Top](#)

 See Also

[IGraphicsBody Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMeshBody Interface](#)

IGraphicsBody Interface Methods

[See Also](#)

	SolidWorks.Interop.sldworks Namespace : IGraphicsBody Interface
---	---

For a list of all members of this type, see [IGraphicsBody members](#).

 Public Methods

Name	Description
 GetBody	Gets the body associated with this graphics body.
 GetFacetCount	Gets the number of facets on this graphics body.
 GetFacets	Gets the facets on this graphics body.
 SurfaceFromMesh	Creates a surface from the selected facets in this graphics body.

[Top](#)

 See Also

[IGraphicsBody Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMeshBody Interface](#)

GetBody Method (IGraphicsBody)

[See Also](#)

 	SolidWorks.Interop.sldworks Namespace > IGraphicsBody Interface : GetBody Method (IGraphicsBody)
---	--

Gets the body associated with this graphics body.

 .NET Syntax

Visual Basic (Declaration)

Function GetBody() As System.Object

Visual Basic (Usage)



```
Dim instance As IGraphicsBody
Dim value As System.Object

value = instance.GetBody()
```

C#

System.object GetBody()

C++/CLI

System.Object^ GetBody();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IBody2](#)

Visual Basic for Applications (VBA) Syntax

See [GraphicsBody::GetBody](#).

Example

See the [IFacet](#) examples.

See Also

[IGraphicsBody Interface](#)
[IGraphicsBody Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetFacetCount Method (IGraphicsBody)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGraphicsBody Interface](#) : GetFacetCount Method (IGraphicsBody)

Gets the number of facets on this graphics body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFacetCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGraphicsBody  
Dim value As System.Integer  
  
value = instance.GetFacetCount()
```

C#

```
System.int GetFacetCount()
```

C++/CLI

```
System.int GetFacetCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of facets

Visual Basic for Applications (VBA) Syntax

See [GraphicsBody::GetFacetCount](#).

Example

See the [IFacet](#) examples.

See Also

[IGraphicsBody Interface](#)
[IGraphicsBody Members](#)

[IGraphicsBody::GetFacets Method \(\)](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFacets Method (IGraphicsBody)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGraphicsBody Interface](#) : GetFacets Method (IGraphicsBody)

Gets the facets on this graphics body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFacets() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGraphicsBody  
Dim value As System.Object
```

```
value = instance.GetFacets()
```

C#

```
System.object GetFacets()
```

C++/CLI

```
System.Object^ GetFacets();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IFacets](#)

Visual Basic for Applications (VBA) Syntax

See [GraphicsBody::GetFacets](#).

Example

See the [IFacet](#) examples.

See Also

[IGraphicsBody Interface](#)
[IGraphicsBody Members](#)
[IGraphicsBody::GetFacetCount Method \(\)](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

SurfaceFromMesh Method (IGraphicsBody)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGraphicsBody Interface](#) : SurfaceFromMesh Method (IGraphicsBody)

Creates a surface from the selected facets in this graphics body.

.NET Syntax

```

Function SurfaceFromMesh( _
    ByVal SurfaceType As System.Integer, _
    ByVal FacetTolerance As System.Double, _
    ByVal ExtendSurfaceSize As System.Double, _
    ByRef ErrorCode As System.Integer _
) As System.Object

```



```

Dim instance As IGraphicsBody
Dim SurfaceType As System.Integer
Dim FacetTolerance As System.Double
Dim ExtendSurfaceSize As System.Double
Dim ErrorCode As System.Integer
Dim value As System.Object

value = instance.SurfaceFromMesh(SurfaceType, FacetTolerance, ExtendSurfaceSize, ErrorCode)

```

```

System.object SurfaceFromMesh(
    System.int SurfaceType,
    System.double FacetTolerance,
    System.double ExtendSurfaceSize,
    out System.int ErrorCode
)

```

```

System.Object^ SurfaceFromMesh(
    & System.int SurfaceType,
    & System.double FacetTolerance,
    & System.double ExtendSurfaceSize,
    & [Out] System.int ErrorCode
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceType

Type of surface to create as defined by [swSurfaceTypes_e](#) (see **Remarks**)

FacetTolerance

0.0 (High tolerance) <= Percent facet selection <= 100.0 (Low tolerance)

ExtendSurfaceSize

Extension in meters of the surface body to fit the selected facets and geometric shape

ErrorCode

0 if successful, -1 if not

Return Value

[IBody2](#)

Visual Basic for Applications (VBA) Syntax

See [GraphicsBody::SurfaceFromMesh](#).

Example

See the [IBody2::ConvertToMeshBody](#) examples.

Remarks

This method:

- is analogous to the **Insert menu > Mesh > Surface From Mesh** functionality in the user interface.
- creates a **Surface-From-MeshX** feature in the FeatureManager design tree.

Before calling this method, select as many facets in the graphics body as are needed to create the surface type.

SurfaceType can be one of only swSurfaceTypes_e.:

- PLANE_TYPE
- SPHERE_TYPE
- CYLINDER_TYPE
- CONE_TYPE

For more information, see the **SOLIDWORKS Design user-interface help > Parts and Features > Graphics Mesh and Mesh BREP Bodies : Creating a Surface from Mesh Feature** topic.

See Also

[IGraphicsBody Interface](#)
[IGraphicsBody Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

IGroundPlaneFeatureData (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGroundPlaneFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGroundPlaneFeatureData Interface

Allows access to a ground plane feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IGroundPlaneFeatureData
```

Visual Basic (Usage)



```
Dim instance As IGroundPlaneFeatureData
```

C#

```
public interface IGroundPlaneFeatureData
```

C++/CLI

```
public interface class IGroundPlaneFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [GroundPlaneFeatureData](#).

Example

'VBA

'=====

'1. Ensure that the specified model exists.
'2. Run the macro.
'3. Creates **Ground Plane1** in the Ground Planes folder of
' the FeatureManager design tree and modifies it.
'=====

```
Dim swApp As SldWorks.SldWorks  
Dim featmgr As SldWorks.FeatureManager  
Dim swGPData As SldWorks.GroundPlaneFeatureData  
Dim featdata As SldWorks.GroundPlaneFeatureData  
Dim ent As SldWorks.Face2  
Dim feat As SldWorks.Feature  
Dim Part As SldWorks.ModelDoc2  
Dim swAssembly As SldWorks.AssemblyDoc  
Dim boolstatus As Boolean  
Dim longstatus As Long, longwarnings As Long  
Option Explicit
```

```
Sub main()
```

```
    Set swApp = Application.SldWorks
```

```
    Set Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2019\samples\tutorial\api\assem2.sldasm", 2  
0, "", longstatus, longwarnings)
```

```
    Set swAssembly = Part  
    swApp.ActivateDoc2 "assem2.sldasm", False, longstatus  
    Set Part = swApp.ActiveDoc  
    Set featmgr = Part.FeatureManager
```

```
    boolstatus = Part.Extension.SelectByRay(-0.124913770805165, 0.118854842937424, 7.02747343940473E-02, -0.620586476260429,  
-0.617739368127351, -0.482980846978724, 3.09726603088958E-03, 2, True, 0, 0)
```

```
    Set ent = Part.SelectionManager.GetSelectedObject6(1, -1)  
    Set swGPData = featmgr.CreateDefinition(swFmGroundPlane)
```

```
swGPData.PlanarEntity = ent

Set feat = featmgr.CreateFeature(swGPData)
Part.ClearSelection2 True
Set featdata = feat.GetDefinition()
boolstatus = Part.Extension.SelectByRay(0.119119826446138, 5.16103810082313E-03, 0.183494239718641, -0.620586476260429,
-0.617739368127351, -0.482980846978724, 3.09726603088958E-03, 2, False, 0, 0)
Set ent = Part.SelectionManager.GetSelectedObject6(1, -1)
featdata.AccessSelections Part, Nothing

featdata.PlanarEntity = ent
feat.ModifyDefinition featdata, Part, Nothing
```

End Sub

 Example

[Create Ground Plane \(C#\)](#)

 Remarks

Use this interface to insert a ground plane at each level of a plant assembly. Ground planes are used as reference geometry when inserting published assets at each level.

Call [IAssemblyDoc::ActivateGroundPlane](#) to activate the ground plane on the level where you want to insert a published asset; the asset's ground face snaps to the assembly's activated ground plane. Components with magnetic mates snap only to components that are also on the active ground plane.

Only one ground plane can be active at a given time in each assembly configuration.

See **Large Assemblies > Facility Layout** in the SOLIDWORKS Design user-interface help.

 Accessors

- [IFeature::GetDefinition](#)
- [IFeatureManager::CreateDefinition](#)

 Access Diagram

[GroundPlaneFeatureData](#)

 See Also

- [IGroundPlaneFeatureData Members](#)
- [SolidWorks.Interop.sldworks Namespace](#)

IGroundPlaneFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGroundPlaneFeatureData Interface

For a list of all members of this type, see [IGroundPlaneFeatureData members](#).

 Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this ground plane feature.
 ReleaseSelectionAccess	Releases access to the selections that define this ground plane feature.

[Top](#)

 See Also

- [IGroundPlaneFeatureData Interface](#)
- [SolidWorks.Interop.sldworks Namespace](#)

IGroundPlaneFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGroundPlaneFeatureData Interface

The following tables list the members exposed by [IGroundPlaneFeatureData](#).

 Public Properties

Name	Description
 PlanarEntity	Gets or sets the planar entity for this ground plane feature.
 ReverseDirection	Gets or sets whether to reverse direction of alignment in this ground plane feature.

[Top](#)

 Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this ground plane feature.
 ReleaseSelectionAccess	Releases access to the selections that define this ground plane feature.

[Top](#)

 See Also

[IGroundPlaneFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IGroundPlaneFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGroundPlaneFeatureData Interface

For a list of all members of this type, see [IGroundPlaneFeatureData members](#).

 Public Properties

Name	Description
 PlanarEntity	Gets or sets the planar entity for this ground plane feature.
 ReverseDirection	Gets or sets whether to reverse direction of alignment in this ground plane feature.

[Top](#)

 See Also

[IGroundPlaneFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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AccessSelections Method (IGroundPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGroundPlaneFeatureData Interface](#) : AccessSelections Method (IGroundPlaneFeatureData)

Gains access to the selections that define this ground plane feature.

 .NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections(  
    ByVal TopDoc As System.Object,   
    ByVal Component As System.Object   
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IGroundPlaneFeatureData  
Dim TopDoc As System.Object
```



```
Dim Component As System.Object
Dim value As System.Boolean

value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(
    System.object TopDoc,
    System.object Component
)
```

C++/CLI

```
System.bool AccessSelections(
    & System.Object^ TopDoc,
    & System.Object^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [GroundPlaneFeatureData::AccessSelections](#).

Example

See the [IGroundPlaneFeatureData](#) example.

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is Nothing or null
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you modified the feature
- [IGroundPlaneFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[IGroundPlaneFeatureData Interface](#)
[IGroundPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

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PlanarEntity Property (IGroundPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGroundPlaneFeatureData Interface](#) : PlanarEntity Property (IGroundPlaneFeatureData)

Gets or sets the planar entity for this ground plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Property PlanarEntity As System.Object
```

```
Dim instance As IGroundPlaneFeatureData
Dim value As System.Object

instance.PlanarEntity = value
value = instance.PlanarEntity
```

C#

```
System.Object PlanarEntity {get; set;}
```

C++/CLI

```
property System.Object^ PlanarEntity {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Planar [face](#) or [plane](#)

Visual Basic for Applications (VBA) Syntax

See [GroundPlaneFeatureData.PlanarEntity](#).

Example

See the [IGroundPlaneFeatureData](#) example.

See Also

[IGroundPlaneFeatureData Interface](#)
[IGroundPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

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ReleaseSelectionAccess Method (IGroundPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGroundPlaneFeatureData Interface](#) : ReleaseSelectionAccess Method (IGroundPlaneFeatureData)

Releases access to the selections that define this ground plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)

```
Dim instance As IGroundPlaneFeatureData

instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [GroundPlaneFeatureData::ReleaseSelectionAccess](#).

Remarks

[IGroundPlaneFeatureData::AccessSelections](#) puts the model into a rollback state to allow access to the selections that define this ground plane feature.

Use this method to restore the rollback state if you did not modify the ground plane feature, or use [IFeature::ModifyDefinition](#) if you did modify the ground plane feature.

See Also

[IGroundPlaneFeatureData Interface](#)
[IGroundPlaneFeatureData Members](#)

Availability

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SOLIDWORKS API Help

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ReverseDirection Property (IGroundPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGroundPlaneFeatureData Interface](#) : ReverseDirection Property (IGroundPlaneFeatureData)

Gets or sets whether to reverse direction of alignment in this ground plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Property ReverseDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGroundPlaneFeatureData  
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

C#

```
System.bool ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse the alignment of assembly faces relative to the ground plane, false to not

Visual Basic for Applications (VBA) Syntax

See [GroundPlaneFeatureData::ReverseDirection](#).

See Also

[IGroundPlaneFeatureData Interface](#)
[IGroundPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0